

Data Analytics Capstone

For this capstone project, you will apply the data analytics theories, methods, and skills you have learned throughout the curriculum to a real-world scenario. You have two pathways for your capstone project: securing a sponsor who provides a genuine data analytics challenge, or selecting an existing dataset with clear real-world applications. Your role will be to transform this input into a data-driven solution that demonstrates your ability to collect, process, analyze data, and produce compelling insights through visualization and presentation.

Requirements

This capstone project consists of three main deliverable components:

1. The Comprehensive Analytics Report

Create a detailed written report that documents the complete story of your project from start to finish. This report must include:

- Introduction of the business challenge, data opportunity, or research question
- Explanation of your data collection/selection and preparation approach
- Documentation of your analytical process and methodology
- Presentation of key findings and insights
- Discussion of recommendations and potential next steps or real-world applications

Artifact Inclusion: The report must include artifacts of the analytical process such as data collection plans, cleaning documentation, statistical analysis outputs, methodology explanations, and any supporting materials that demonstrate the depth and legitimacy of your work.

2. The Presentation Slide Deck

Develop a compelling visual presentation that effectively communicates your project insights to a professional audience. This slide deck must include:

- Clear storytelling that guides the audience through your analytical journey
- Professional visual design appropriate for business presentation

- **Data Visualization Component:** Present findings using appropriate tools such as Tableau, Python libraries (Matplotlib, Seaborn), or other visualization platforms that effectively communicate insights
- Actionable recommendations and next steps
- Supporting charts, graphs, and visual elements that enhance understanding

3. Delivering the Presentation on Demo Day

Live Presentation Format: Present your capstone project live via Zoom during the designated "Demo Day."

Duration: At least 10-15 minutes to clearly communicate the business problem, your analytical process, key findings, data visualizations, and actionable recommendations.

Audience Consideration: Your audience will include peers, instructors, and invited guests. Assume no prior knowledge of your project—provide sufficient context for anyone to understand the business scope, data challenge or opportunity, analytical approach, and impact of your proposed insights.